Lunar tidal potential measured on a coastal–inland optical-clock baseline

AUTHOR-REVIEW CONSOLIDATION — ALL RESULTS PENDING RECALCULATION. This manuscript follows the v29 lunar-versus-solar-day narrative and processing route. R01–R18 are unresolved numerical fields. Historical source figures are labelled previews; no fit, simulation or raw-data processing was performed for this version. Outcome wording is provisional until recalculation. New Fig. 3d–f and Extended Data Fig. 7 are pending. All spectral ordinates use a linear scale. Reference verification limits are recorded in REFERENCE_CHECK.md.

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Optical clocks measure differences in gravitational potential through the gravitational redshift. Detecting tidal changes with distant clocks requires distinguishing astronomical signals from laboratory variations that repeat with the solar day. Here we measure the lunar semidiurnal tidal potential difference between strontium and ytterbium clocks in coastal Shanghai and inland Wuhan, separated by 692 km and connected by a 1,350-km fibre link. Observations across several campaigns sample the changing phase of the lunar tide relative to local time. A joint fit of astronomical tidal groups gives a lunar response of [R03: estimate and uncertainty pending] relative to the combined solid-Earth and ocean-loading prediction. We quantify its sensitivity to daily disturbances with either a common phase or independent campaign phases, and assess environmental records, transmission monitoring and alternative processing. These measurements test the recovery of lunar tidal potential changes from an interrupted clock comparison, with attribution limited by the observed phase coverage, measurement noise and physical-model uncertainty.

Introduction

Comparing two optical clocks measures the difference in gravitational potential at their locations, the basis of chronometric levelling.[1–4] This observable complements gravitational acceleration and displacement measured by gravimeters and satellite position-

ing. Static potential differences have been measured with laboratory and transportable clocks,[5–9] down to the gradient across a millimetre-scale atomic sample,[10] and remote comparisons have extended these measurements through fibre networks.[11,12] Extending clock comparisons to time-varying potential would provide a means of observing the dynamics of the gravity field. Tides offer a predictable signal for this purpose, with spatially varying contributions from astronomical forcing, solid-Earth deformation and ocean loading.[13–15]

The difficulty is that the tides share their periods with variations in laboratory conditions. The diurnal constituents K_1 and P_1 lie close to one cycle per solar day, while S_2 has exactly two cycles per solar day. During a short observing campaign, these signals can be difficult to distinguish from disturbances that repeat at the same local time. The principal lunar semidiurnal tide M_2 provides a different temporal signature: its maximum occurs about 50 min later each day. Observations over a sufficiently broad range of lunar–solar phases can therefore distinguish lunar variation from some solar-day disturbances. The strength of this distinction depends on the actual observation times, noise and assumed stability of the disturbance phase.

Here we investigate the lunar tidal potential difference between Shanghai and Wuhan on a coastal–inland baseline. Ocean loading contributes to the predicted signal because the two stations respond differently to the redistribution of seawater. We analyse observations between September 2025 and August 2026, including the summer record used in an earlier absolute frequency-ratio measurement.[12] We jointly estimate lunar semidiurnal, neighbouring semidiurnal and diurnal responses, with astronomical phases and relative constituent amplitudes fixed within each group. The lunar group is the primary observable; the complete tidal waveform and other groups provide context and cross-checks. We quantify the estimator's response to solar-day disturbances, examine environmental and transmission records, and compare alternative processing of the shared observations.

Results

A coastal–inland tidal potential difference

The Shanghai Sr and Wuhan Yb clocks are linked through a phase-stabilized fibre network routed through Hefei (Fig. 1a).[12] Shanghai lies near the East China Sea and Wuhan inland. The comparison senses the difference in the potential at the two clocks, including their motion with the deforming Earth. Our prediction combines an IERS-based solid-Earth tide with the full ocean-loading potential calculated from FES2022 ocean elevations.[16–19] The latter includes direct attraction by seawater, the induced potential of the deformed Earth and the vertical displacement of the clock.

For M_2, the predicted differential solid-Earth and ocean-loading phasors have amplitudes [R02: solid amplitude] and [R02: loading amplitude] and a phase difference of [R02: phase difference] (Fig. 1b). Their vector sum gives [R02: combined amplitude]. The corresponding loading contribution on the actual sampling is quantified after removing each segment's mean. These model quantities describe the physical signal tested by the comparison; they do not constitute separate experimental detections of solid-Earth and ocean-loading effects.

The record is organized into the five v29 campaigns: September–October 2025, January, April and May 2026, and June–August 2026. The selected observations comprise [R01: retained segment count] segments and [R01: accepted hours] h, with the relation to operating groups and the eight observation periods documented in the sample ledger (Fig. 1c). These grouping levels serve different purposes and are not interchangeable. Individual counter samples and overlapping filtered observations are temporally correlated measurements, not independent experimental repeats.

Separating lunar variation from the solar day

We apply the same 1,200-s triangular filter at 600-s steps to the data and the predicted tide. Each segment receives an independent constant frequency offset. The astronomical model is separated into the lunar semidiurnal group M_2+N_2, the neighbouring group S_2+K_2 and the diurnal group. A joint fit estimates one response coefficient per group, fixing the phases and relative amplitudes within each group. The primary fit contains no empirical daily terms; additional fits quantify its sensitivity to such terms.

The lunar response is A_L = [R03: estimate ± calibrated statistical standard error], with [R03: confidence interval and confidence level] (Figs. 2 and 3a). Unity denotes the model prediction. The neighbouring and diurnal responses are [R04: estimates and standard errors], and the covariance between lunar and neighbouring responses is [R04: covariance and correlation] (Fig. 3b). These coefficients describe responses to groups of constituents; they do not establish separate measurements of N_2 or K_2. An in-phase and quadrature fit provides a check of lunar timing, giving [R05: Cartesian coefficients and covariance]. Amplitude and phase summaries will be reported only where the fitted signal supports a meaningful phase estimate.

The summer record samples a changing lunar–solar phase (Fig. 1d). Its contribution to the lunar information is [R06: projected information share], calculated after projecting out the other tidal groups and segment constants. The lunar estimates from summer alone and from the earlier campaigns are [R06: subset estimates and uncertainties]. Campaign-resolved estimates and their consistency test are reported in Extended Data Table 3. Their interpretation depends on both phase coverage and measurement precision; elapsed calendar span alone does not determine the information in the record.

Figure 2 presents the summer lunar variation after accounting for segment offsets and the other groups. Its compressed time axis marks removed gaps explicitly, while Extended Data Fig. 7 retains the actual UTC dates and all interruptions. Phase-bin uncertainty will include refitting of nuisance terms or equivalent full covariance propagation through the filtering and projection. The historical v29 conditional error bars are not treated as this revised uncertainty calculation.

Frequency-domain checks of the semidiurnal response

The full-record and April spectra in v29 are supplemented by local scans around M_2 and S_2, single-component subtraction and conditional spectra (Fig. 3c–f). All scans use the primary selected record, the same triangular filtering and the actual observation times after segment-offset projection. Spectral ordinates are linear, with a common scale for direct subtraction and conditional comparisons. Their peak locations and normalized powers are [R07/R18: pending]. To test the observation window, a single M_2 or S_2 constituent is passed through that complete processing operator. Its power at the other principal frequency and the local sidelobe pattern quantify predicted cross-frequency

leakage [R18: transfer ratios and frequency-dependent responses].

We compare the observed spectrum with spectra obtained after subtracting only the jointly estimated M_2 component or only the jointly estimated S_2 component. The removed time series is formed before applying the same spectral operator; powers are not subtracted. Agreement between a disappearing feature and the injected single-constituent response would support a leakage interpretation. It cannot alone prove that the entire observed feature is leakage, because fitted subtraction, interference and genuine correlated components can also change spectral power.

Conditional scans fit the competing principal frequency's free sine and cosine terms together with the segment constants while scanning the target frequency. Their retained structure is [R18: conditional results and uncertainty]. Separation of structure near M_2 and S_2 does not distinguish an S_2 tide from a laboratory disturbance at exactly 12 h. Spectra, amplitude fits and phase checks use the same observations; they complement one another and do not accumulate independent detection significance. Reduction after subtraction of the full fitted tide remains an auxiliary diagnostic, not an instrumental signal-to-noise ratio.

Whole-template response and relation to earlier work

A single coefficient fitted to the full tidal prediction gives $A_T =$ [R08: estimate and calibrated standard error]; the semidiurnal-band response is [R08: estimate and standard error]. These fits use the same observations as the grouped analysis and are not combined as independent measurements. Campaign-resolved results, the April time-domain comparison and the associated spectral diagnostics are given in Extended Data Figs. 3 and 4.

The summer measurements overlap the absolute Yb/Sr ratio study of Hou and colleagues.[12] The present study extends the observing record and estimates the lunar group separately. Table 1 distinguishes the earlier reported estimator, the present filtered analysis and a separate reprocessing with non-overlapping 600-s means. Differences in sample selection, reference levels, weighting and uncertainty calculation must be retained when comparing these estimates. The reprocessing is a check using shared experimental records, not an independent experiment, and its numerical results remain [R14: pending matched reprocessing].

Sensitivity to daily laboratory effects

We quantify the response to a sinusoidal disturbance of prescribed amplitude rather than assuming that all solar-day effects are absent. For a 10^{-18} disturbance with a phase common to the entire record, the largest change in A_L is [R09: 24-h response] at 24 h and [R09: 12-h response] at 12 h. Allowing a different phase in each campaign gives worst-case changes of [R09: independent-phase responses] (Fig. 4b). These are conditional transfer sensitivities for the specified disturbance models. They do not bound arbitrary nonstationary laboratory effects.

We also fit common 24-h and 12-h terms, campaign-specific terms, temperature-phase-constrained terms and measured temperature regressors (Fig. 4a). The corresponding lunar coefficients and uncertainties are [R10: sensitivity estimates]. Independently phased 12-h terms can compete strongly with astronomical semidiurnal templates within short campaigns. Their effect must therefore be judged from both the change in the lunar estimate and its uncertainty. The neighbouring S_2+K_2 response is not given equal evidential status to the primary lunar result.

Both clocks apply blackbody-shift corrections. The Sr chamber-temperature record tests residual effects associated with that correction over its actual overlap with the clock data. The overlap, fitted coefficient and uncertainty are [R11: matched environmental-control results]. Air-temperature regressions describe association with external conditions; they do not establish a causal temperature correction or exclude unmeasured laboratory influences.

Cross-checks and physical-model sensitivity

We test prediction by fitting the tidal response outside each observation period and evaluating it on the held-out period while fitting only its segment offsets. The change in residual sum of squares is [R12: per-period and pooled prediction results]. Residual resampling, operating-group bootstrap and restricted period-level wild bootstrap provide complementary checks of statistical uncertainty and null behaviour under their respective assumptions (Methods). No result of these checks is inferred from a low residual alone.

The fibre round-trip monitor provides a control of transmission effects. With the clock sample, monitor normalization, valid-time rules and matching fixed, adding the monitor

changes the lunar and whole-template coefficients by [R13: paired coefficient changes and uncertainties]. The final matched exposure is [R13: hours]. The historical v29 and v86 coverage claims refer to different processing routes; neither is adopted until a timestamp- and member-level reconciliation accounts for their difference.

We retain v29's physical-model tests: radial-displacement validation of the loading convolution, the S_2 discrepancy between model products and sensitivity to the near-coastal ocean mask (Extended Data Fig. 6). The mask alternatives change A_L by [R16: signed changes and envelope]. This is a physical-model sensitivity distinct from statistical error. Its magnitude and direction affect how agreement with unity is interpreted; it is not dismissed solely because it falls inside a statistical error bar or added in quadrature without a justified probability model.

Transient selection and reference-level choices are tested with the same primary processing (Fig. 4a and Extended Data Fig. 5). The fraction of samples removed, selection-invariance checks and changes with the transients retained are [R15: selection and robustness results]. Measured residual correlation, time-based covariance estimates and resampling within continuous runs will additionally be evaluated on this filtered record, adapting the verification methods described in v86 (Methods). Those checks are pending; their v86 numerical outcomes are not transferred to this analysis.

Discussion

The scientific question is whether a distant optical-clock comparison can distinguish lunar tidal potential changes from variation associated with the solar day. The lunar tide provides a test because its phase moves relative to local time. The primary result is the response of the lunar semidiurnal group, interpreted jointly with its phase, the actual observing window and quantified sensitivity to specified daily disturbances. The final numerical conclusion remains contingent on the common-route recalculation identified above.

Broad lunar–solar phase coverage can reduce confusion with a disturbance whose phase is stable in local time. It does not remove the effects of finite noise, limited observing time or a disturbance that changes between campaigns. Our main fit uses astronomical templates without empirical daily terms, while supplementary fits and transfer calculations assess this modelling choice. In particular, campaign-specific 12-h terms set a

more demanding attribution test than a common-phase disturbance. Phase coverage, noise level, effective observing time and nuisance-model assumptions together determine the recoverable information.

Ocean loading makes the coastal–inland geometry useful for testing the combined tidal potential. The solid-Earth and loading contributions are partly aligned, however, so agreement with their sum does not independently identify the loading component. The separate loading estimate [R16: estimate and uncertainty] and the coastal-mask sensitivity delimit this interpretation. Longer observations across seasons, improved clock stability and baselines with different relative loading phases could strengthen such tests.

The experiment develops a potential-sensitive measurement alongside established gravity and displacement observations. Its demonstrated scope should be assessed from the recalculated lunar response and its uncertainty, rather than from unrelated static clock accuracy or spectral peak suppression. A related observation with a European optical-clock network provides complementary context.[20] The present contribution is the explicit assessment of lunar–solar-day separation on a coastal–inland baseline, using an extended record whose relationship to the earlier frequency-ratio study is stated directly.

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Main figures and table

Figure status. Panels below reproduce the supplied v29 artwork solely to review scientific intent and layout. Every plotted value and error bar is historical and awaits regeneration. Revised captions specify the final intended calculation and identify differences from the previews.

V29 SOURCE PREVIEW - NOT RECALCULATED

All plotted values and error bars are historical; replacement pending.

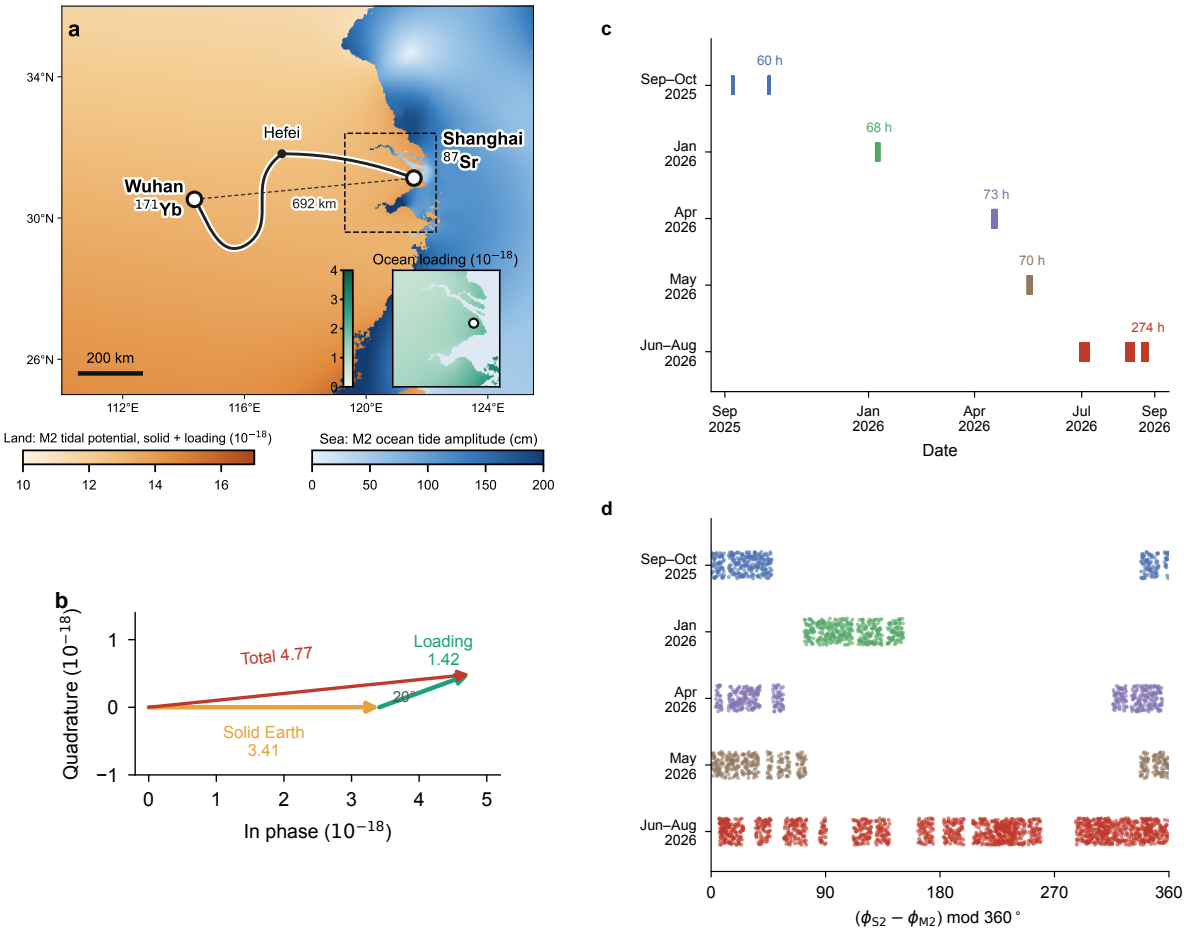


Fig. 1 | The predicted tide and lunar-solar phase coverage on a coastal–inland baseline. **a,** Shanghai and Wuhan clock sites and the fibre route through Hefei; land colours represent the predicted M_2 effective tidal potential in fractional-frequency units and sea colours the ocean-tide elevation. The inset shows ocean loading near Shanghai. **b,** Differential M_2 phasors for the solid-Earth tide, ocean loading and their sum. These are predictions, not independent measurements of the components. **c,** Actual sampling by campaign. **d,** Relative S_2-M_2 astronomical phase, including nodal corrections, at those times. Recalculate all phasors, sample locations, exposure labels and phase coverage using the final retained-member ledger (R01, R02, R06). Preserve this figure's geometry and phase-coverage purpose.

V29 SOURCE PREVIEW - NOT RECALCULATED

All plotted values and error bars are historical; replacement pending.

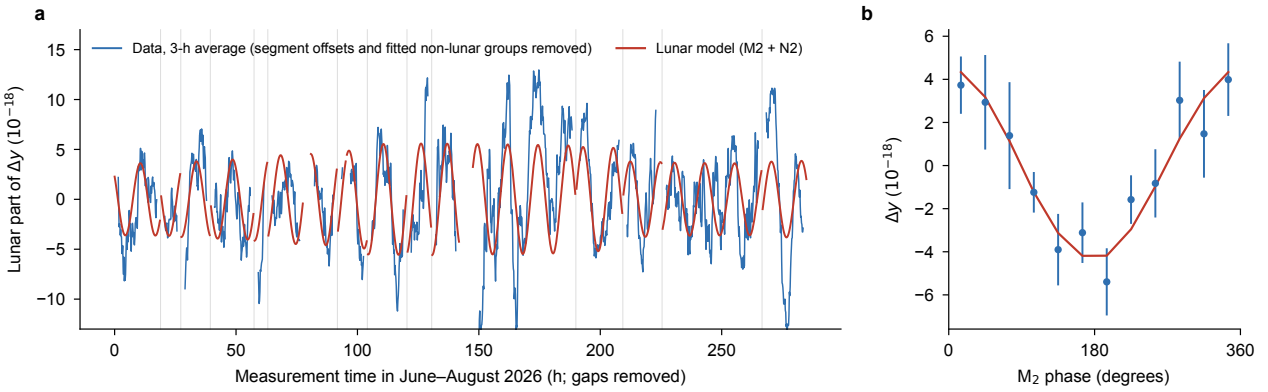


Fig. 2 | Lunar variation in the summer clock comparison. a, Three-hour display averages after accounting for segment constants and fitted non-lunar groups, with the lunar prediction filtered and averaged on the same support. The horizontal axis is compressed measurement time: gaps longer than 6 h are removed and explicitly marked. It must not be read as uninterrupted observation. Extended Data Fig. 7 will show the same data against actual UTC with every gap retained. **b,** Phase-folded data and prediction in 30° M_2 bins. The historical preview holds fitted nuisance terms fixed when estimating jackknife error; the replacement must refit these terms in each replicate or propagate their full covariance. Bin support counts, uncertainty definition and correlations must accompany the regenerated panel (R01, R03, R05).

V29 SOURCE PREVIEW - NOT RECALCULATED

All plotted values and error bars are historical; replacement pending.

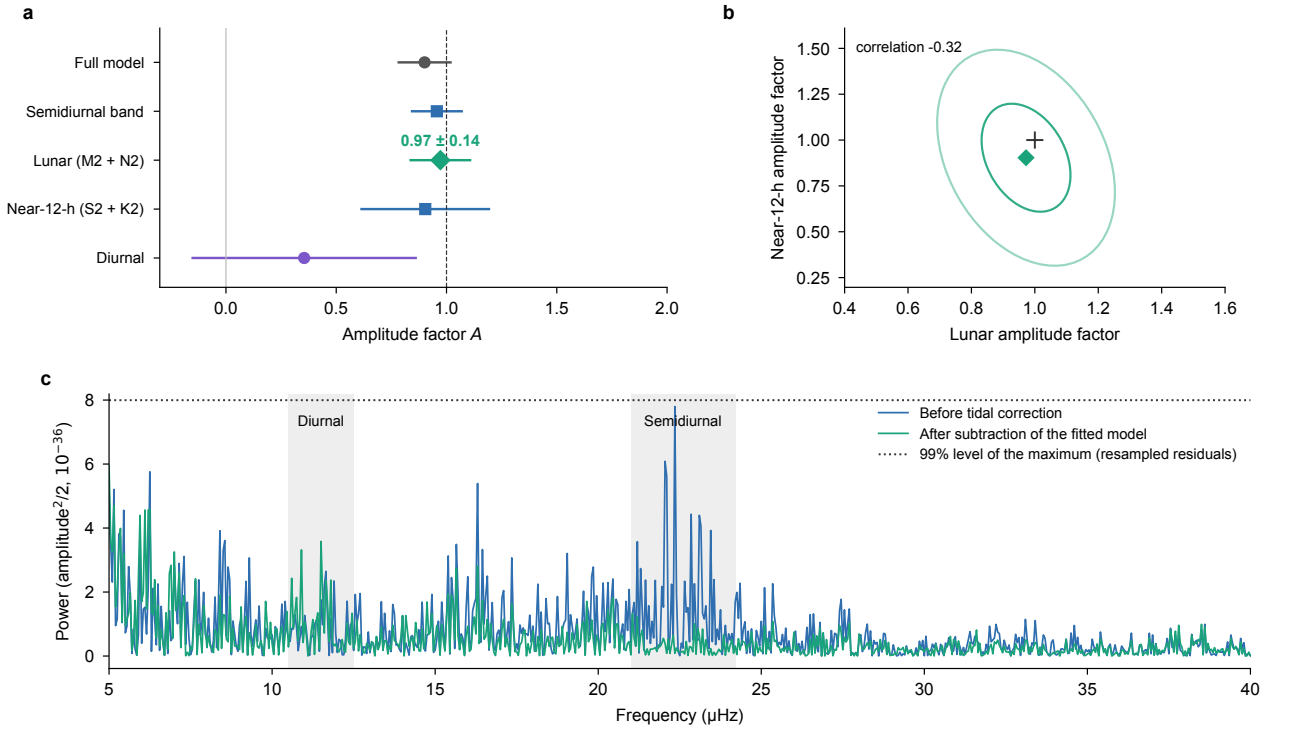


Fig. 3 | Tidal responses and frequency-domain checks. **a,** Full-template, semidiurnal-band and grouped responses; these share observations. **b,** Joint lunar–neighbouring uncertainty, with covariance and joint-region interpretation specified. Marginal standard-error radii are not automatically joint confidence levels. **c,** Full-record sinusoidal power before and after fitted full-template subtraction. The ordinate is linear, starting at zero; its normalization is amplitude squared divided by two, not density per μHz. All direct spectral comparisons share the same linear scale. **d–f, pending new panels: d,** Local M_2/S_2 spectra of the same observations, after removing only the jointly fitted M_2 pair and after removing only the jointly fitted S_2 pair. **e,** Single-constituent injection responses through the actual selection, filter and offset projection, with cross-frequency power ratios. **f,** Conditional scans controlling the competing principal frequency's free sine and cosine terms. All panels use the same primary processing and fixed normalization (R03, R04, R07, R08, R18). The preview contains only v29 panels a–c; no d–f results have yet been generated. Peak suppression alone does not prove leakage. Frequency separation does not resolve S_2 from an exactly 12-h laboratory disturbance, and these panels supply no independent significance to add to the amplitude fit.

V29 SOURCE PREVIEW - NOT RECALCULATED

All plotted values and error bars are historical; replacement pending.

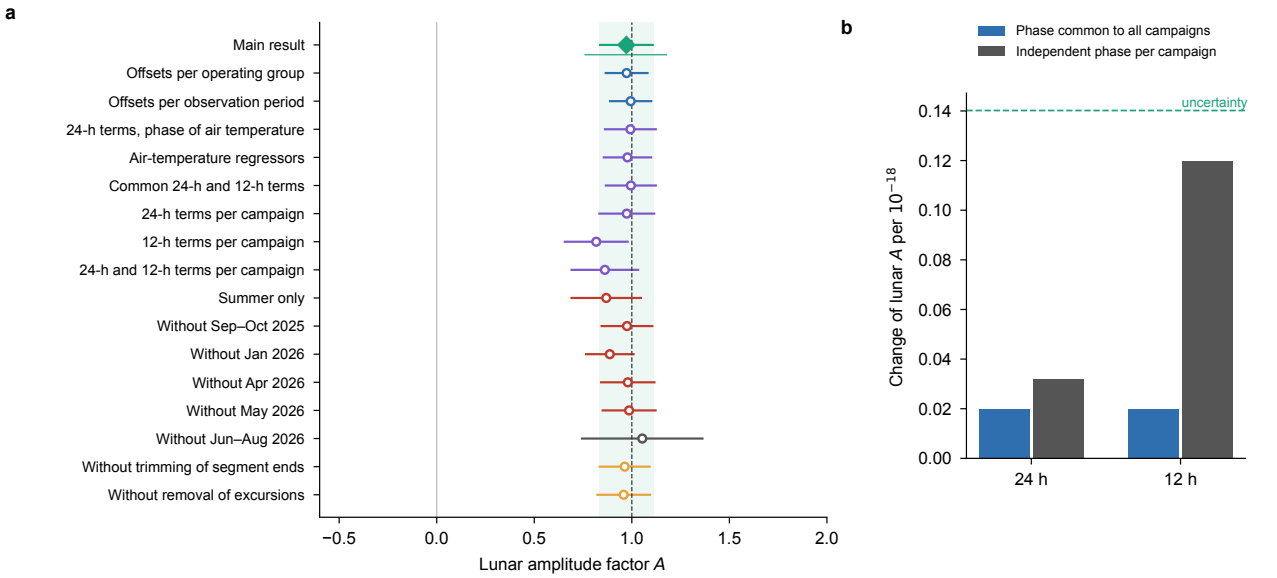


Fig. 4 | Sensitivity of the lunar estimate to analysis choices and solar-day disturbances. **a,** Primary estimate, alternative reference levels, temperature-related and sinusoidal nuisance models, subsets, campaign deletions and selection variants. Fits share observations. Recalculate every point and its uncertainty under the final processing; retain the group-bootstrap interval as a separately defined interval. **b,** Largest coefficient change for a prescribed 10^{-18} sinusoid at 24 h or 12 h, for a phase common to the record and for phases independent between campaigns. Both assumptions must remain visible. The reference statistical uncertainty and all bars await recalculation (R03, R06, R09, R10, R15). This figure tests specified disturbance models and does not bound arbitrary time-varying laboratory errors.

Table 1 | Related analyses of shared observations. All current-study numerical cells are pending. The earlier study is a published/preprint comparator, not a value to tune the present fit towards. Verify its exact reported number and definition directly from the cited source before filling the first row. Do not combine rows as independent evidence.

Analysis	Sample and estimator	Response	Uncertainty and status
Hou et al.[12]	Summer record; original segment-wise estimator and original tidal model	R14: source-check pending	Original study's reported definition
Primary lunar analysis	v29 selected record; triangular filter; joint groups and segment offsets	R03: pending	Calibrated statistical error and interval
Whole-template analysis	Same v29 record and filter; single physical coefficient	R08: pending	Calibrated statistical error
Non-overlapping-mean reprocessing	Separately specified 2026 sample, clock corrections and calendar reference levels	R14: pending	Design-specific LS/delete-day/delete-period comparison
Lunar reprocessing with daily terms	Same reprocessing sample; its explicit nuisance design	R14: pending	Design-specific uncertainty; not transferred to primary fit

Methods

Analysis status and processing route

This author-review consolidation specifies the v29 processing route as the primary analysis. All experimental estimates, sample counts, uncertainties, validation outcomes and data-dependent graphics require recalculation. Bracketed R identifiers link to the numerical register. Descriptions below specify the intended processing unless explicitly attributed to an earlier study; no new fit, simulation or raw-data processing was performed for this consolidation. Additional v86 verification methods are identified separately and must be adapted to this route before they support a result.

Clocks, frequency chain and timestamps

The Shanghai Sr and Wuhan Yb lattice clocks, frequency combs and fibre comparison are described in the earlier frequency-ratio study.[12] Independent clock evaluations provide the instrument context.[21,22] The model coordinates inherited from v29 are 31.130° N, 121.570° E for Shanghai and 30.529° N, 114.357° E for Wuhan. Approximate station heights, coordinate frame and observation-specific clock settings must be bound to the source metadata before recalculation. Published systematic clock evaluations must not be substituted for the statistical uncertainty of the tidal comparison. Related field-fibre demonstrations provide context, rather than measurements of this comparison's transmission noise.[23,24]

The observable is the fractional variation of Wuhan Yb relative to Shanghai Sr. With positive geodetic potential W,

$$y(t) = -\frac{\delta W_{\mathrm{Wuhan}}(t) - \delta W_{\mathrm{Shanghai}}(t)}{c^2}.$$

The 2025 counter records a divided beat, so its deviations require a factor of two relative to the undivided chain; the frequency sign and scale must be established from the frequency-chain records, not chosen to improve a tidal fit. Static potential and frequency offsets are absorbed by segment constants. Absolute frequency-ratio arithmetic, where needed for the conversion, must retain adequate decimal precision.

The v29 route reconstructs each file's time axis from its first timestamp and nominal one-second sample index. This requires a check of missing records, counter-gate con-

ventions and file continuity against the original timestamps. Beijing time is converted to UTC once, and the astronomical time convention is recorded separately. Synchronization accuracy is not inferred from timezone labels. The proposed timing sensitivity shifts the model by ±10 min; resulting coefficient changes are [R17: timing sensitivity].

Selection and matched filtering

The experiment-team interval table defines the candidate segments. The v29 sequence excludes beat deviations exceeding 10 Hz from the segment median, retains the longest uninterrupted run and trims leading and trailing samples using the stated 1% peak-to-peak rule. The precise endpoint algorithm, threshold reference and order of operations must be bound to executable code; prose alone is insufficient to recover a unique implementation. No selection threshold is adjusted to bring a response closer to unity.

To identify smaller transients, v29 forms 300-s averages, compares them with a running median over ±30 min, and excludes averages exceeding six robust standard deviations of these differences within a segment, together with their neighbours. The robust-scale definition, minimum local support and number of neighbouring blocks are implementation details requiring explicit resolution. This criterion is inherited from an analysis already informed by the data; it is not newly described as preregistered. Selection sensitivity includes 60- and 300-s blocks, thresholds from four to eight, retained transients and omitted endpoint trimming. Model injections with factors -1, 1 and 3 test whether the selected membership changes; their outcomes remain [R15].

Retained one-second observations and every model component undergo identical 1,200-s triangular filtering with a 600-s step. Window centring, endpoint weights and treatment of missing members must be recorded. Filters cannot bridge a discontinuity merely because output points lie on a regular grid. The number of filtered points is distinct from accepted one-second exposure and elapsed span. A member ledger will map every filtered row to source file, row identity, timestamp, weight, segment, operating group, campaign and observation period. Overlap between neighbouring windows induces correlation even when counter noise is white.

Solid-Earth tidal potential

The v29 solid-tide model directly combines astronomical potential, induced potential and clock displacement:

$$\delta W_{\text{solid}} = \sum_{m=0}^{2} (1 + k_{2m}) V_{2m} + (1 + k_3) V_3 - g u_r.$$

Here V_{2m} are the lunar and solar degree-two tide-generating potentials separated by order, V_3 is the lunar degree-three contribution and u_r is radial displacement. The v29 constants are $k_{20} = 0.29525$, $k_{22} = 0.29801$ and $k_3 = 0.093$. The order-one response uses the IERS resonance relation constituent by constituent; the radial displacement uses the IERS algorithm with frequency and latitude dependence of the displacement response.[16] Ephemeris identity, tidal system, Earth frame, Love-number convention and time scale must accompany the regenerated templates. This formulation is already a direct-potential calculation; a criticism of a displacement-to-potential approximation in another version does not automatically apply to it.

Ocean-loading potential

The full loading calculation follows the v29 convolution,

$$\delta W_{\text{load}}(t) = \int_{\text{ocean}} \rho_w \zeta(\mathbf{r}', t) G_W(\psi) \, \mathrm{d}A,$$

$$G_W(\psi) = \frac{ga}{M_E} \sum_n (1 + k'_n - h'_n) P_n(\cos \psi).$$

Here ζ is ocean-tide elevation, ρ_w is seawater density, ψ is angular separation, a and M_E are Earth's radius and mass, and h'_n, k'_n are load Love numbers.[19] The prescription uses FES2022 elevations on a 1/30° grid and $\rho_w = 1,025\,\mathrm{kg\,m^{-3}}$.[17,18] PREM load Love numbers through degree 696 follow Han and Wahr and the gravity-toolkit distribution.[27,28] Above that degree the displacement response is smoothly joined to the stated Boussinesq limit −6.22, with a Kummer-transformed sum to degree 40,000. The degree-zero treatment, potential datum and geocentre convention require explicit code-level documentation.

The convolution uses the full grid within 3° of each site and 8-by-8 cell blocks farther away. Ocean-mean removal enforces the adopted mass-conservation convention. Eight constituents, Q_1, O_1, P_1, K_1, N_2, M_2, S_2 and K_2, are synthesized with astronomical arguments and nodal corrections. Model version, land mask, interpolation and normalization must be fixed across sites and tests.

The radial displacement from the same convolution is compared with the FES2022 load-tide grid. This checks the displacement calculation and convolution but does not independently validate every potential term. The S_2 difference between products is retained and its effect on all fitted groups is recalculated. The coastal-mask test fills land cells within 0.3° and 0.6° of the coastline in a 4°-by-5° region centred on Shanghai with nearest-ocean values. These are sensitivity scenarios, not corrected observations or a probability distribution for coastline error. Report signed response changes and the envelope separately from statistical errors, alongside alternative-grid and numerical-convergence checks [R16].

Response models and inference

The whole-template fit is $y_i = c_{s(i)} + A_T s_T(t_i) + \epsilon_i$. The primary grouped fit is

$$y_i = c_{s(i)} + A_L s_L(t_i) + A_S s_S(t_i) + A_D s_D(t_i) + r(t_i) + \epsilon_i,$$

where s_L contains M_2+N_2, s_S contains S_2+K_2, s_D contains the four diurnal constituents and r is the fixed minor-constituent remainder. All terms are filtered like the observations. The band fit instead assigns independent coefficients to semidiurnal and diurnal bands. Separate constants represent segment reference levels, not physical jumps established by the fit. The v29 main route is least squares; no v86 noise weights are silently imported. Exact row weights must be confirmed from the corresponding v29 implementation before execution.

Scaled least-squares covariance and a delete-day jackknife are recorded. The jackknife deletes a specified calendar day's observations and refits the complete model, including nuisance coefficients; the day boundary must be fixed in the execution configuration. The original v29 description does not uniquely establish that boundary. For the principal lunar, whole-template and semidiurnal coefficients, v29 additionally circularly shifts

residuals within each continuous filtered run and adds them to a unit-amplitude model. Two independent sets of 200 realizations and the component standard errors are to be retained, with the reported selection rule fixed in advance. Any use of the largest uncertainty must be stated explicitly. Artificial adjacency introduced at a circular shift's wrap point is a limitation, assessed by the continuous-run block checks below.

Operating-group and segment bootstraps use 1,000 draws and assume independence between their respective resampling units. They are distinct diagnostics because segments within an operating group need not be independent. A phase check jointly estimates in-phase and quarter-cycle-shifted templates; Cartesian coefficients and covariance are primary. A near-zero amplitude cannot support a precise phase. For phase-folded displays, each jackknife replicate refits the non-lunar and segment terms, or an equivalent full covariance propagation is used. Error bars from nuisance coefficients held fixed must remain labelled as conditional if shown for historical comparison.

The v29 injection suite includes lunar amplitudes zero and unity, with no daily disturbance, temperature-phase daily terms, common 24-h plus 12-h terms, and summer-derived daily terms. Its design uses 300 realizations per injection scenario, 5,000 per null-disturbance scenario for tail behaviour, and 5,000 per specified coverage scenario. These are proposed replication settings, not counts executed here. Report biases, empirical standard deviations, interval coverage and binomial uncertainty. A zero exceedance count yields a finite upper confidence bound, not a zero probability. The exact one- or two-sided tail convention must accompany the reported count. The period-level restricted wild bootstrap enumerates whole-period residual sign patterns; its two-sided definition and null restriction must be documented. No multiple-testing-adjusted detection claim is inferred from examining several diagnostic fits.

Residual and uncertainty checks adapted from v86

On the recalculated v29 residuals, inspect serial correlation and increment statistics within actual continuous runs, retaining missing intervals. Frequency-stability definitions follow Riley.[26] Compare the primary standard errors with delete-day and delete-period estimates, time-based Bartlett covariance estimates and moving-block residual resampling. The proposed v86-inspired block durations are 1, 3, 6 and 12 h with 2,000 realizations per feasible setting. Use actual elapsed time and do not join separate runs.

Short runs limit which blocks are possible; the eligibility rule, retained support and effect of the triangular filter must be reported for each setting.

These checks use the same v29 sample, templates, reference levels and estimator. Bandwidth choice, residual centring, finite-sample correction and random seeds must be recorded before execution. They assess error stability under stated dependence assumptions; resampling the same residuals is not an independent validation of the true noise process. No v86 numerical error, coverage rate or model-comparison result is imported into this manuscript.

Daily-disturbance projection and environmental records

For angular frequency ω, a daily disturbance is $B\cos(\omega t - \varphi)$. If h is the lunar row of the actual least-squares operator, its largest coefficient shift for a common phase is

$$G_{\mathrm{common}} = B\sqrt{(h^{\mathsf{T}}\cos\omega t)^2 + (h^{\mathsf{T}}\sin\omega t)^2}.$$

For independent campaign phases, sum these norms over campaign-restricted vectors. Apply the same filtering and membership to the disturbance vectors as to the data. These bounds assume the prescribed amplitude and sinusoidal form; they do not cover arbitrary amplitude modulation. Report both phase assumptions for 24-h and 12-h disturbances [R09].

Additional regressions use common or campaign-specific sine and cosine terms. ERA5 hourly near-surface air temperatures supply campaign daily phases or running-mean anomalies as exploratory regressors.[25] The Sr chamber record averages 20 PT100 sensors sampled every 20 s, as described in v29. Its blackbody sensitivity and temperature reference must be documented from the clock correction model, and the actual correction status established for the retained samples. The 30 June transient is analysed with and without exclusion; the rule and affected membership remain visible. Overlap durations, daily coefficients and regression changes are all pending [R11].

Fibre monitoring, held-out prediction and alternative processing

The v29 round-trip monitor is normalized by the optical carrier frequency without division by two and filtered like the clock ratio. It excludes deviations exceeding 10 Hz from

its median and tests an additional 300-s-block criterion of 5×10^{-16}. Matching must preserve source-channel identity, timestamp convention, counter validity and one-second membership. Compare regressions with and without the monitor on exactly the same clock support, and use paired deletion or resampling for coefficient-change uncertainty. A ledger must reconcile the historical 474-h v29 coverage and 246.6-h v86 matched coverage by selection stage. Neither duration is a new result of this consolidation.

Held-out prediction fits the physical response on the other periods, then estimates only local segment offsets on the held-out data. Report per-period and pooled residual sums of squares on unchanged test support. The separate reprocessing uses non-overlapping 600-s means, clock systematic corrections and its own fixed calendar reference levels; it must remain separately labelled in Table 1. Its white, Ornstein–Uhlenbeck, Student-t and moving-average simulations evaluate that design, not the primary filtered estimator. The segment-wise estimator of the earlier study, and the v29 Kalman check with a segment-reset offset random walk, are likewise auxiliary analyses. Their settings and results require distinct provenance [R14].

Spectral displays and reproducibility

Spectra fit sinusoids at actual observation times after the stated offset projection and use the v29 amplitude-squared-over-two normalization. This is not interchangeable with power spectral density per unit frequency. Show the observation window and state the search grid, frequency band and any frequencywise or maximum-over-band resampling threshold. The v29 main-spectrum caption specifies 200 residual series over 5–40 µHz, while its April check uses 1,000 series over 15–50 µHz; keep these distinct and resolve the exact generation scripts. Filtered sample count, duration, model residuals and every figure value must come from one versioned output manifest.

The additional local, leakage and conditional scans are specified in SPECTRAL_CHECK_PROTOCOL (R18). Both the injected constituents and all sinusoidal design columns pass through the same raw-member selection and triangular filter as the measurements. Compare the M_2-only and S_2-only power responses on the fixed search grid. For each direction report cross-frequency power divided by the injected constituent's own-frequency power, with the phase convention stated and phase dependence explored. A raw timestamp-window Fourier transform alone is not the full estimator's leakage response.

For single-component subtraction, estimate free sine/cosine pairs for M_2 and S_2 jointly with segment constants in an explicitly labelled spectral diagnostic. Remove only the fitted M_2 pair or only the fitted S_2 pair from the time series while keeping all row identities, weights, offset treatment and normalization fixed. Also show fixed-physical-template subtraction as a separate model-dependent diagnostic where available. Neither diagnostic changes the main grouped astronomical amplitude estimator. Conditional scans fit the competing pair simultaneously with the scanned pair and offsets, equivalently projecting both the observations and scanned design onto the complement of the nuisance design. They are not obtained by subtracting a spectrum. Any resampling repeats fitting and subtraction in each realization; the family of scanned frequencies and a maximum-over-band threshold must be specified before interpreting a probability. No v86 leakage percentage or spectrum is substituted for this pending recalculation.

Numerical confidence statements are two-sided unless explicitly defined otherwise. The lunar response is the primary target; model alternatives, subgroups, spectra and environmental controls are sensitivity or exploratory analyses. Report the testing family and any multiplicity adjustment if inferential claims are made across those checks. Software versions, input checksums, random seeds and run logs accompany the future numerical release. The present package records document-generation software and source-code provenance only.

Data availability

Original counter records and experimental logs remain in the controlled research archive. The final retained-member ledger, derived observations, templates and figure-source tables are pending recalculation. Access terms and a permanent identifier require author confirmation. No public data deposit is claimed.

Code availability

The local author-review package contains a versioned snapshot of the available program library, its provenance inventory, the v29 processing specification and manuscript-build scripts. The complete executable chain that generated the supplied v29 PDF has not yet been bound to a unique source revision. The package therefore supports source review and document reproduction but does not claim completed numerical reproduction of

v29. A final analysis release and its access conditions will be specified after recalculation and author review.

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Author contributions

[Contribution statement pending confirmation by all authors.]

Competing interests

[Author declaration pending.]

AI-assisted preparation

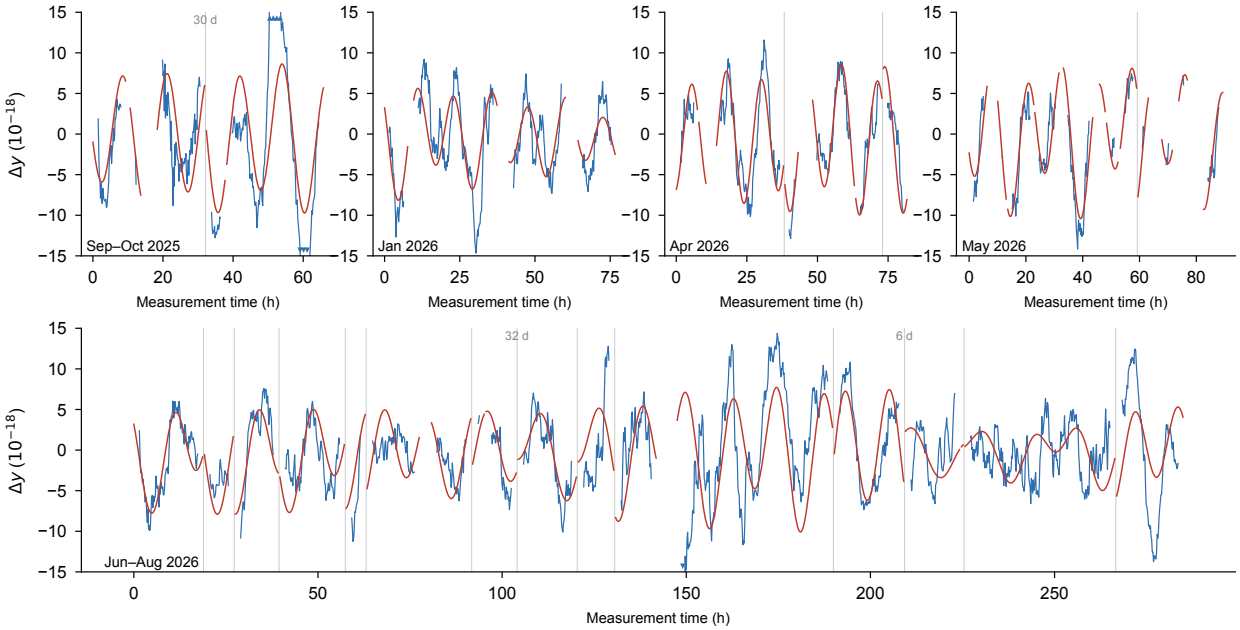
OpenAI Codex assisted this consolidation of manuscript versions, preparation of the cover-letter draft, source-code packaging and document checks. This consolidation did not recalculate experimental results. The authors remain responsible for verification of the analysis, interpretation and approval of the final manuscript. Descriptions of earlier AI-assisted work require author review.

Extended Data

Status. Figures 1–6 are labelled historical previews from v29; final curves and uncertainty bars are pending. Figure 7 and all tables are defined below without fabricated values. The six original figure topics and three table topics are retained; the new UTC panel is the additional validation requested in this consolidation.

V29 SOURCE PREVIEW - NOT RECALCULATED

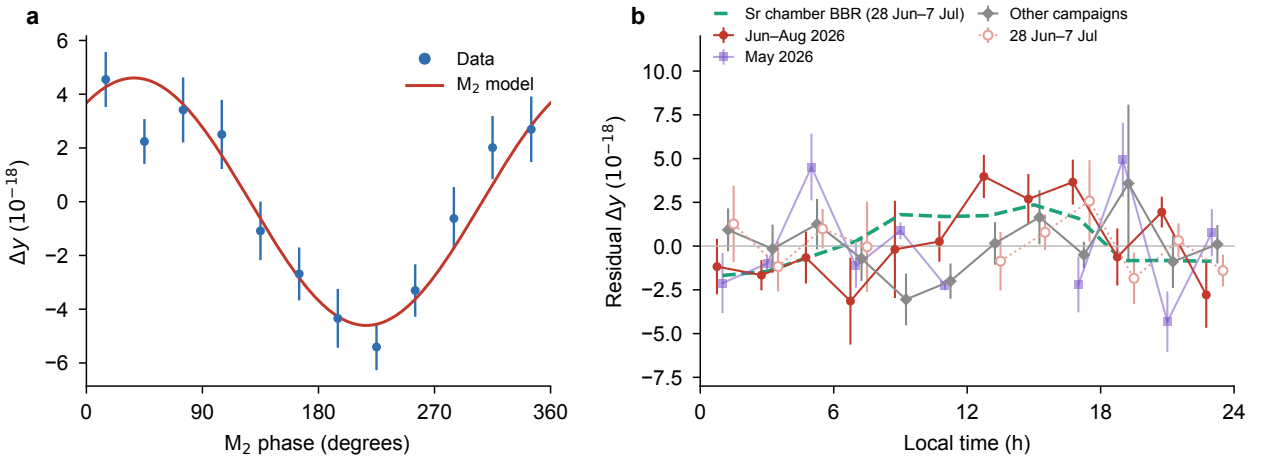
All plotted values and error bars are historical; replacement pending.



Extended Data Fig. 1 | Campaign time series. Three-hour averages after segment-offset subtraction, with the matched model. The original compressed axes remove gaps longer than 6 h and mark omitted intervals. The source preview clips the vertical range and uses triangles for excursions; the regenerated figure should show the full range or supply a full-range companion and exact overflow counts. Actual dates and all gaps are retained in Extended Data Fig. 7. All data and support counts: R01 and R08.

V29 SOURCE PREVIEW - NOT RECALCULATED

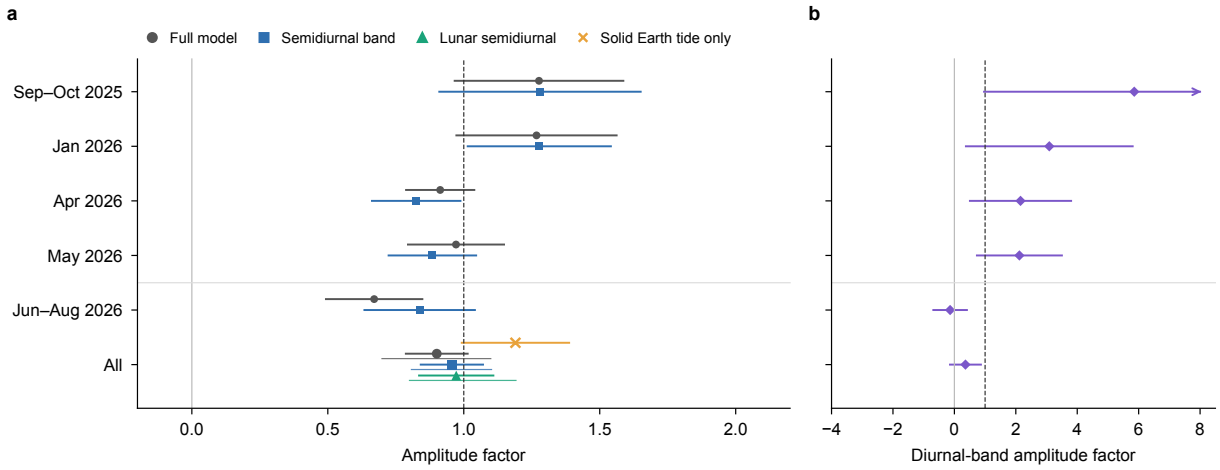
All plotted values and error bars are historical; replacement pending.



Extended Data Fig. 2 | Tidal-phase and local-time displays. a, M_2 phase fold after accounting for the other model terms. **b,** Local-time residual profiles by campaign and over the chamber-temperature overlap, compared with the uncorrected blackbody shift that the real-time correction is intended to remove. The temperature transient is shown in declared included/excluded variants. Final errors must propagate refitting of nuisance terms; sparse-bin exclusions and per-bin day counts must be stated (R05, R11).

V29 SOURCE PREVIEW - NOT RECALCULATED

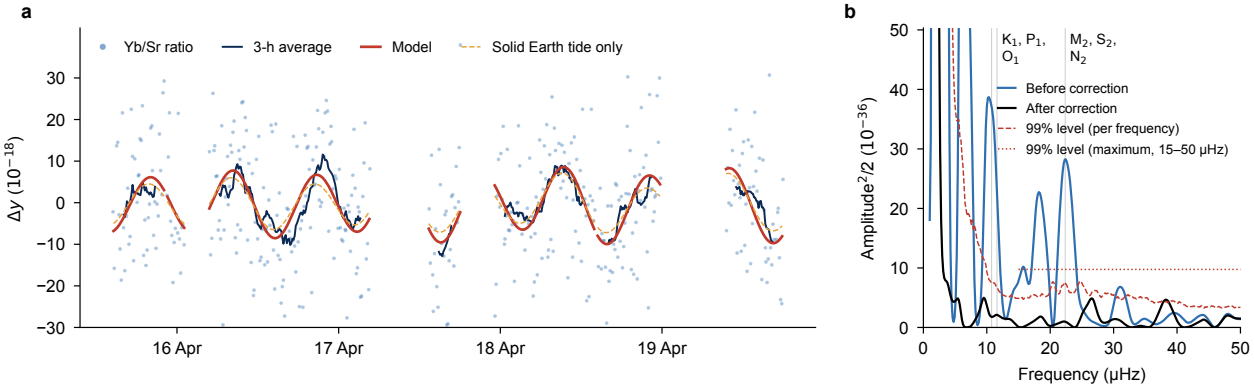
All plotted values and error bars are historical; replacement pending.



Extended Data Fig. 3 | Campaign-resolved tidal responses. **a,** Complete-template and semidiurnal-band coefficients, with the all-data lunar and solid-only comparisons separately labelled. **b,** Diurnal-band coefficients. Regenerate every interval with its stated error method and show full uncertainty ranges. Report the sample and resampling-unit count per campaign. Shared-observation fits do not provide independent replications (R01, R04, R06, R08, R16).

V29 SOURCE PREVIEW - NOT RECALCULATED

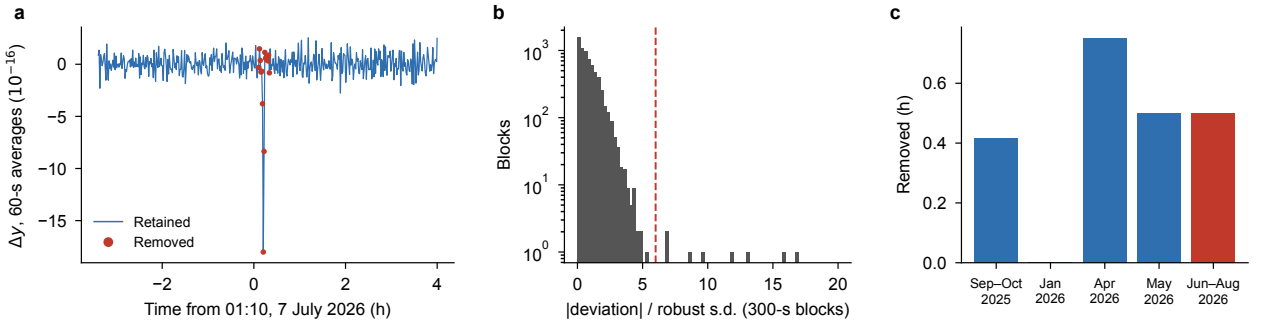
All plotted values and error bars are historical; replacement pending.



Extended Data Fig. 4 | April observations and spectral diagnostic. **a,** Filtered clock observations, display averages and matched predictions against actual UTC. **b,** Sinusoidal power before and after model subtraction, with frequencywise and maximum-over-band resampling thresholds distinguished. Document the frequency band, number of residual realizations and treatment of low-frequency power outside the display. The April data alone do not separately resolve all constituents in each band (R07, R08).

V29 SOURCE PREVIEW - NOT RECALCULATED

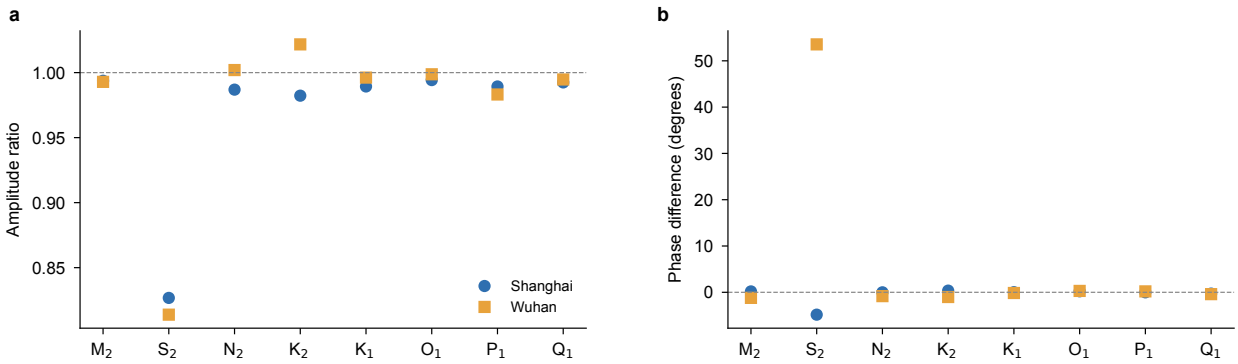
All plotted values and error bars are historical; replacement pending.



Extended Data Fig. 5 | Transient-selection audit. a, The July excursion at a stated display averaging time, with retained and removed one-second members identified. **b,** Distribution of robust deviations and the declared threshold. **c,** Removed exposure by campaign. Regenerate from the complete member ledger; report robust-scale definition, neighbour removal, source identities and selection changes under tidal injection. No exclusion is justified by improved agreement with the desired response (R15).

V29 SOURCE PREVIEW - NOT RECALCULATED

All plotted values and error bars are historical; replacement pending.



Extended Data Fig. 6 | Displacement-component validation of ocean loading. Amplitude ratios and phase differences between the loading convolution's radial displacement and the specified FES2022 load-tide product. Retain the S_2 discrepancy and document product release and reference frame. Agreement of displacement alone does not independently validate attraction and induced-potential terms. The coastal-mask response changes are reported separately as physical-model sensitivity (R16).

Extended Data Fig. 7 | Actual-date view of the lunar record — pending generation. Planned panels show (a) all campaigns on a real UTC axis and (b) the summer observations used in Fig. 2, with gaps left blank and the same data/model projection. Show raw filtered points with suitable transparency, display averages on explicitly defined windows and per-window support. This is a required new figure, not an image already produced. It will be generated only after the R01 member ledger and the R03/R05 refits are available.

Extended Data Table 1 | Noise-model simulations of the separate reprocessing. R14; no simulations run in this consolidation. Preserve the distinction between this design and the primary v29 estimator. Each scenario requires realized count, seed, coefficient bias, empirical SD and coverage of the nominal 95% interval, including binomial uncertainty. OU denotes Ornstein–Uhlenbeck noise.

Scenario inherited from v29	Full-template bias / coverage	Lunar bias / coverage
White noise	Pending	Pending
White + OU, 1 h	Pending	Pending
White + OU, 12 h	Pending	Pending
White + OU, 3 d	Pending	Pending
White + period constants	Pending	Pending
Student-t, df 7, MA(1)	Pending	Pending
Student-t, df 5, MA(1)	Pending	Pending
Student-t, df 7, MA(1) + OU	Pending	Pending

Extended Data Table 2 | Injection checks of the primary lunar estimator. R03/R10; no simulations run. For each row and injected lunar amplitude 0 and 1, report bias, empirical SD, median estimated standard error, one-standard-error coverage with a 95% binomial interval, 1.96-standard-error coverage and the fraction exceeding three estimated errors. Keep conditional simulation assumptions explicit.

Daily disturbance	Injected amplitudes	Bias, SD, estimated error, coverage and exceedance
None	0; 1	Pending for each amplitude
Campaign 24 h, air-temperature phase	0; 1	Pending for each amplitude
Common 24 h + 12 h	0; 1	Pending for each amplitude
Summer-derived 24 h + 12 h	0; 1	Pending for each amplitude

Extended Data Table 3 | Campaign-level daily terms and tidal responses. R01/R06/R10. For each campaign report observed day count, 24-h amplitude and phase with uncertainty, 12-h amplitude with uncertainty, whole-template response and lunar-group response. Weak daily amplitudes can be positively biased and may not support meaningful phase estimates.

Campaign	Observed days	24-h amplitude / maximum local time	12-h amplitude	Full / lunar response
September–October 2025	Pending	Pending	Pending	Pending
January 2026	Pending	Pending	Pending	Pending
April 2026	Pending	Pending	Pending	Pending
May 2026	Pending	Pending	Pending	Pending
June–August 2026	Pending	Pending	Pending	Pending

Residual-dependence comparisons, physical-model sensitivities and the transmission-support reconciliation will be supplied as machine-readable supplementary tables R13, R16 and R17. Their eventual placement and figure/table allocation should be checked against the journal's current display limits after the numerical outputs exist.